Ritika Sibal

ritikasibal2@gmail.com | (734) 780-9410 | ritikasibal.github.io

EDUCATION

Master of Science in Biology, Boston University — May 2025 | GPA: 3.96/4.0

Thesis: *Investigating orangutan locomotion using computer vision;* **Awards:** Merit Scholarship, Denton Award for Best Thesis

Bachelor of Science in Computer Engineering, University of Michigan — Jun 2020 | GPA: 3.5/4.0

RESEARCH EXPERIENCE

Primate Ecology and Conservation Lab — **Aug 2023 – Present**

Graduate Research Student — Boston, MA

- Developed two novel computer vision models for automated classification of primate joint positions (98% accuracy).
  - Labeled high-resolution video data; performed data preprocessing (including image augmentation), and assessed quality.
  - Created a secondary ML model (SVM, 99% accuracy) to automatically predict body posture from joint coordinates.
  - Coded end-to-end data pipelines for automatic extraction and analysis of kinematic data (from model predictions).
  - Analyzed terrestrial gait in primates using camera trap footage–first quantitative description of wild orangutan gait.
- Managed and extracted from long-term datasets (using SQL); modeled population developmental trends (GLMMs, Bayesian).
- Training a novel species recognition model for AI-based biodiversity analysis using trail camera trap footage.
- Collaborating with international teams (*Oxford*) to build an orangutan facial recognition model, targeting field deployment.
- Designing prototype thermal imaging computer vision tool to expand non-invasive monitoring approaches in challenging habitats.

Deep Optical Perception Lab — **Aug 2016 – Jun 2020**

Undergraduate Research Student — Ann Arbor, MI

- Built a hybrid multi-modal model (97% accuracy) to predict dolphin swimming patterns from wearable sensor + video data.
  - Deployed predictive tool to a dolphin sanctuary to monitor individual animal welfare and longitudinal behavior.
- Designed an autonomous aquatic survey vehicle; integrated GPS and onboard sensors for aquatic habitat assessment.
  - Collected underwater imagery and habitat data; created interactive bathymetry maps and evaluated habitat quality.

WORK EXPERIENCE

Atmosic Technologies — **May 2022 – Jul 2023**

Embedded Software Engineer — Boston, MA

- Developed radio communication drivers (802.15.4) for a custom chip, enabling low-power IoT applications.
- Improved software acceptance and regression testing by creating custom debugging tools and automated CI/CD testing pipelines

Apple — **Aug 2020 – May 2022**

Firmware Automation Engineer — Cupertino, CA

- Led 2021 MacBook release: coordinated across engineering, manufacturing, and customer service teams.
- Designed new automation frameworks combining Python scripting, Embedded C, CAD, and peripheral management (motors, linear actuators, signal generators) improving QA/QC and removing manual effort. Recognized for technical innovation.

PERSONAL PROJECTS

Custom Goat Wheelchair (2022): Designed and built a 3D-printed goat wheelchair, published workflow online; featured on CBS.

Autonomous Battery Charging for Wildlife Drones (2020): Engineered an autonomous wildlife drone flight and charging system; implemented automated GPS waypoint navigation, obstacle avoidance, and end-to-end precision landing.

PUBLICATIONS (selected)

- Sibal, R. et al. "DeepHutan: A novel computer vision model for primate pose estimation." (Submitted, July 2025)
- Sibal, R. et al. "Bidirectional LSTM Recurrent Neural Network Plus Hidden Markov Model For Wearable Sensor Based Dynamic State Estimation." (Dynamic Systems and Control, May 2019)

SKILLS

Programming: Python, R (including tidyverse, RMarkdown), MATLAB, C++/C, SQL, Version Control (GitHub),
Machine Learning: Supervised/Unsupervised Learning, Classical ML (K-means, Random Forest, SVM, PCA), Computer Vision, Large Language Models, Deep Learning (CNNs, RNNs, GANs).
Statistical Modeling: Generalized Additive and Generalized Linear Mixed Models (GAMMs, GLMMs), Species Distribution Modeling, Habitat Modeling, Passive Acoustic Monitoring, Bayesian Statistics.
Other Technologies: Raven, ArcGIS, Oracle, CAD (Autodesk Fusion 360), Docker, CI/CD (Jenkins), GPS, Bluetooth, RFID.